ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

COMPUTER HARDWARE AND SOFTWARE

PAPER 1 · PRACTICE A · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 3 Computer Hardware and Software paper 1 · practice a.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Accept answers in the student's own words if the meaning is correct, and other correct examples. In conversions, give method marks for correct working.

SECTION A

QUESTION 1

1.1 1.1.1 D ✓ 1.1.2 B ✓ 1.1.3 D ✓ 1.1.4 B ✓ 1.1.5 C ✓ 1.1.6 A ✓ 1.1.7 B ✓ 1.1.8 D ✓ 1.1.9 A ✓ 1.1.10 C ✓ (10)

1.2 1.2.1 True ✓ 1.2.2 False ✓ 1.2.3 False ✓ 1.2.4 True ✓ 1.2.5 True ✓ 1.2.6 True ✓ 1.2.7 True ✓ 1.2.8 True ✓ 1.2.9 False ✓ 1.2.10 True ✓ (10)

1.3 1.3.1 B ✓ 1.3.2 H ✓ 1.3.3 E ✓ 1.3.4 J ✓ 1.3.5 C ✓ 1.3.6 G ✓ 1.3.7 D ✓ 1.3.8 F ✓ 1.3.9 A ✓ 1.3.10 I ✓ (10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2

2.1 Any FIVE: CPU socket ✓; RAM (DIMM) slots ✓; chipset ✓; PCIe expansion slots ✓; SATA connectors ✓; BIOS / UEFI chip; CMOS battery; M.2 slot; power connectors (24-pin ATX); back-panel ports (USB, audio, network) (5)

2.2 Fetch: the CPU (control unit) gets the next instruction from RAM using the address in the program counter ✓✓; decode: the control unit works out what the instruction means and which operation is needed ✓✓; execute: the ALU carries out the instruction and the result is stored in a register or memory ✓✓. (6)

2.3 Speed: SSD much faster (no moving parts, faster boot and loading) ✓ – HDD slower (spinning platters, moving head) ✓; Durability: SSD more durable / shock-resistant ✓ – HDD can be damaged if dropped ✓; Cost: SSD more expensive per GB ✓ – HDD cheaper per GB, larger capacities ✓ (6)

2.4 More programs and data can be kept in RAM at the same time ✓, so the computer uses virtual memory (the slower drive) less ✓, and switching between programs is faster ✓. (3)

[20]

QUESTION 3

3.1 $173 = 128 + 32 + 8 + 4 + 1$ ✓ (or repeated division by 2 with remainders) ✓ → 1010 1101 ✓ (3)

3.2 $1101 = \text{D}, 0110 = 6$ ✓ → D6 ✓ (2)

3.3 $2 \times 16 + 15$ ✓ $= 47$ ✓ (2)

3.4 $182 + 59 = 241$ (working) ✓✓ → 1111 0001 ✓ (3)

3.5 ASCII is a code that represents characters as numbers (7 or 8 bits, up to 256 characters) ✓. Unicode uses more bits, so it can represent characters of all the world's languages, symbols and emoji ✓✓. (3)

3.6 Manufacturers count 1 GB as 1 000 000 000 bytes ✓, while Windows counts 1 GB as 1 024 × 1 024 × 1 024 bytes, so the same space shows as a smaller number (500 × 10^9 ÷ 1 073 741 824 ≈ 465) ✓. (2)

[15]

QUESTION 4

4.1 Any FIVE in a logical order ✓✓✓✓✓: check hardware requirements; set the boot order in the BIOS / UEFI to the USB or DVD; boot from the installation media; choose language and enter the product key / accept the licence; partition and format the drive; copy files and install; create a user account and set the time zone; install device drivers; install updates and antivirus (5)

4.2 Any TWO: the software might not run or might run very slowly ✓; to avoid wasting money and time / avoid crashes ✓; to know if hardware must be upgraded (2)

4.3 Any TWO: disk imaging / cloning (install on one, copy the image to all) ✓; network deployment tools (e.g. Windows Deployment Services, group policy) ✓; unattended (silent) installation scripts; package managers (2)

4.4 Open-source software: source code is freely available and may be changed and shared ✓, e.g. Linux, LibreOffice, Python ✓; proprietary software: owned by a company, code is secret and a licence must be bought ✓, e.g. Microsoft Windows / Office, Adobe Photoshop ✓. (4)

4.5 Each student logs in with their own account ✓, and permissions stop students from changing settings or installing software / protect files ✓. (2)

[15]

QUESTION 5

5.1 Any FOUR with sensible frequencies (1 + 1 each): clean dust from fans and vents – every 3 to 6 months ✓✓; run antivirus scans / update definitions – daily or weekly ✓✓; install operating system updates – monthly / when available ✓✓; back up data – daily or weekly ✓✓; check cables and hardware for damage; delete temporary files / disk clean-up – monthly; check the UPS battery (8)

5.2 Identify the problem (gather information from the user) ✓; establish a theory of probable cause ✓; test the theory ✓; establish a plan of action and implement the solution ✓; verify full system functionality ✓; document findings, actions and outcomes ✓ (6)

5.3 (a) Overheating ✓ – clean the fans / heat sink, replace thermal paste, check that fans work ✓ (b) Malware / adware infection ✓ – run an antivirus and anti-malware scan, remove unknown programs and browser extensions ✓ (4)

5.4 Take it to a registered e-waste recycling centre / return to the supplier ✓, because computers contain toxic materials (lead, mercury) that harm the environment, and parts can be recycled ✓; wipe data first. (2)

[20]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE